

L^AT_EX

A Document Preparation System

2019



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L^AT_EX



Outline

Introduction

What is T_EX and L^AT_EX?

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$\text{\LaTeX}$



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What is T_EX?

T_EX is a typesetting system written by Donald E. Knuth, who says in the Preface to his book on T_EX that it is “*intended for the creation of beautiful books — and especially for books that contain a lot of mathematics.*”

T_EX is a macro processor, and offers its users a powerful programming capability. T_EX on its own is a pretty difficult beast to deal with, so Knuth provided a package of macros for use with T_EX called Plain T_EX;

$$\begin{aligned}\frac{\partial \rho}{\partial t} &= \operatorname{div}(\rho \bar{\mathbf{v}}) \\ \rho \frac{D \bar{\mathbf{v}}}{Dt} &= \rho \bar{\mathbf{g}} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) + \delta_{ij} \lambda \operatorname{div} \bar{\mathbf{v}} \right] \\ \rho \frac{De}{Dt} &= \operatorname{div}(k \nabla T)\end{aligned}$$

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What is L^AT_EX?

- L^AT_EX is a T_EX macro package, originally written by Leslie Lamport, that provides a document processing system. L^AT_EX allows markup to describe the structure of a document, so that the user need not think about presentation. By using document classes and add-on packages, the same document can be produced in a variety of different layouts.
- L^AT_EX is for producing scientific and mathematical documents of high typographical quality and is also suitable for producing all sorts of other documents, from simple letters to complete books.
- L^AT_EX is *not* a word processor! Instead, L^AT_EX encourages authors not to worry too much about the appearance of their documents but to concentrate on getting the right content.

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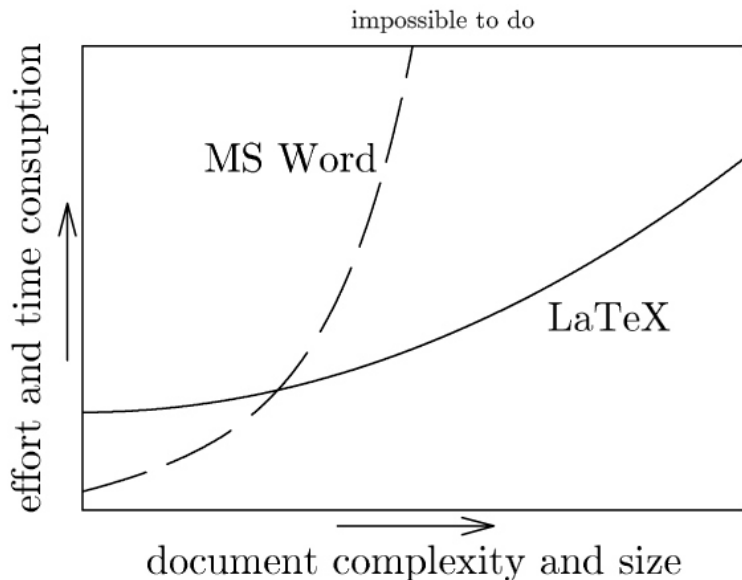
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WYSIWYG vs Markup languages

Word processors (WYSIWYG)

- Microsoft Word, Open Office, etc.
- Have a WYSIWYG interface hiding the markup.
- They're perceived to be easier than $\text{\LaTeX}$, but ...

Markup languages

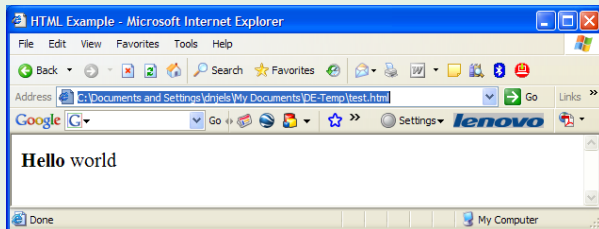
- Written in normal text with markup in a text editor.
- Needs to be interpreted or compiled (programming language)
 - HTML
 - $\text{\LaTeX}$
- Contents separated from format; uses style sheets to change appearance.



HTML Example: test.html

```
<html>
  <head>
    <title>HTML Example</title>
  </head>
  <body>
    <p>
      <b>Hello</b> world
    </p>
  </body>
</html>
```

Output





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```

\documentclass{article}
\begin{document}
  If  $f(x)=3x+7$  and  $g(x)=x+4$  then
  \begin{equation}
    f(x) + g(x) = 4x + 11
  \end{equation}
  and
  \begin{equation}
    f(x)g(x) = 3x^2 + 19x + 28.
  \end{equation}
\end{document}

```

Output

If $f(x) = 3x + 7$ and $g(x) = x + 4$ then

$$f(x) + g(x) = 4x + 11 \tag{1}$$

and

$$f(x)g(x) = 3x^2 + 19x + 28. \tag{2}$$

*Create
and edit
document*

myfile.tex

```
\documentclass{article}
\begin{document}
  Hello world
\end{document}
```

Compile

`>latex myfile.tex`

`>pdflatex myfile.tex`

*View and
print*

myfile.dvi

Hello world

myfile.pdf

Hello world

```
>dvips myfile.dvi
>ps2pdf myfile.ps
```

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Why use $\text{\LaTeX}$?

- Very high quality documents.
- The most sophisticated math typesetting engine available.
- Sophisticated page, paragraph and line breaks with hyphenation.
- Micro typographical ligatures, letters spacing, hanging punctuation, etc.
- Floating figures for optimal page layout with graphics and tables.
- Easy cross referencing of equations, figures, etc. with hyperref links.
- Powerful linkage to bibliography databases, citing mechanisms and bibliography formatting.
- Automatic tables of contents.
- Automatic language settings.
- *Did I mentioned very high quality documents.*

$\text{\LaTeX}$



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Micro typesetting



Kerning

Table, Avant

Table, Avant

Ligatures

fire, office

fire, office

Letterspace

D a n i e

Danie

True small caps

CAPE TOWN

CAPE TOWN

Hanging Punct

derstand anything. So much so
that those who saw him exclaimed:
“What a burden he’ll be to his fa-
ther!” Now when there was any-
thing to be done, the eldest had



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Active chars # \$ % ^ & _ { } \

\ – escape char to activate commands

\$ – activate math

% – comment char

Special chars

\”{e}, {\0} – gives ë, Ø

Inline commands

\rmfamily, \itshape, etc.

Macros

\section{Syntax} – cmdnd with argument

\func[x]{cos} – cmdnd with optional arg

Environments

\begin{itemize}

\item ...

\end{itemize}



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Basic L^AT_EX document layout

```
\documentclass[a4paper,12pt]{article}
```

```
% Preamble -----
```

```
\usepackage{amsmath}
```

```
% Beginning of main document -----
```

```
\begin{document}
```

```
\section{Functions}% Heading ----
```

In equation (`\ref{eq:a}`) the quadratic function of x is shown as

```
\begin{equation}
```

```
f(x)=ax^2 + bx + c
```

```
\label{eq:a}
```

```
\end{equation}
```

It can be shown that $x \geq \sqrt{\alpha}$ and ...

```
\end{document}
```

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1. Functions

In equation (1) the quadratic function of x is shown as

$$f(x) = ax^2 + bx + c \quad (1)$$

It can be shown that $x \geq \sqrt{a}$ and ...

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L^AT_EX classes

```
\documentclass[a4paper,12pt]{article}
```

Specify the basic formatting of the document: Type block, headers, title page, paragraph indents and spacing, etc.

Optional settings for paper size, font size, etc.

- **article, report, book:** Standard L^AT_EX classes
- **memoir, koma:** Other useful specialized classes
- **amsart, elsevier, etc.:** Journal classes
- **usthesis:** Stellenbosch thesis class

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L^AT_EX packages

```
\usepackage{amsmath}
```

There are hundreds of different packages available on CTAN for any thing you can think of. Some useful packages for theses are:

- **amsmath**: Advanced mathematical typesetting
- **hyperref**: Hyper links in PDF documents
- **graphicx**: Inclusion of graphics
- **array**: For tables and arrays
- **siunitx**: Type setting of units and numbers
- **fourier**: A very nice complete font set
-
-

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Main document

```
\begin{document}  
    ....  
\end{document}
```

This is the main contents of the document.

Headings

```
\section{Functions}
```

L^AT_EX has a hierarchy of headings

- **\part**
- **\chapter**
- **\section, \subsection, \subsubsection**
- **\paragraph, \subparagraph**



Paragraph text

In equation (`\ref{eq:a}`) the quadratic function of x is shown.

This is a new paragraph.

A new paragraph is terminated by an open line in the text.

Math

```
\begin{equation}
  f(x)=ax^2 + bx + c
\end{equation}
```

Mathematics can be written inline as $x=3$ or as display math as in example above.

The AMS math package gives a wealth of different ways to format difficult equations and please RTFM!

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In equation (`\ref{eq:a}`) the quadratic function of x is shown

```
\begin{equation}
  f(x)=ax^2 + bx + c
  \label{eq:a}
\end{equation}
```

Output

In equation (1) the quadratic function of x is shown

$$f(x) = ax^2 + bx + c \tag{1}$$

The `\label{...}` and `\ref{...}` commands provide a very powerful way to cross ref all the sectioning levels, equations, figures and tables, etc.

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Enumerate:

```
\begin{enumerate}
  \item The first item
  \item The second item
\end{enumerate}
```

Output

- 1 The first item
- 2 The second item

Itemize:

```
\begin{itemize}
  \item The first item
  \item The second item
\end{itemize}
```

Output

- The first item
- The second item

Description:

```
\begin{description}
  \item[First] item
  \item[Second] item
\end{description}
```

Output

First item
Second item



Figures

```
\begin{figure}
  \includegraphics{myfigure}
  \caption{A beautiful figure}
  \label{fig:1}
\end{figure}
```

Figures can be included from external files (or generated inside $\text{\LaTeX}$). They float to where $\text{\LaTeX}$ find it best.

- $\text{\LaTeX}$: PostScript
- $\text{\PDFLaTeX}$: PDF, PNG, JPEG, MPOST

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Tables

```
\begin{table}
\begin{tabular}{l*{6}{c}r}
\hline
Team                & P & W & D & L & F   & A & Pts \\
\hline
Manchester United & 6 & 4 & 0 & 2 & 10  & 5 & 12 \\
Celtic            & 6 & 3 & 0 & 3 & 8   & 9 & 9 \\
Benfica           & 6 & 2 & 1 & 3 & 7   & 8 & 7 \\
\hline
\end{tabular}
\end{table}
```

Output

Team	P	W	D	L	F	A	Pts
Manchester United	6	4	0	2	10	5	12
Celtic	6	3	0	3	8	9	9
Benfica	6	2	1	3	7	8	7

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The BibTeX companion program makes it possible to link a bibliographic database direct into a LaTeX document.

BibTeX data item

```
@ARTICLE{Ciamarra-2005,  
  author = {Ciamarra, M. and Coniglio, A. and Nicodemi, M.},  
  title  = {Shear instabilities in granular mixtures},  
  journal = {Physical Review Letters},  
  year   = {2005},  
  volume = {94},  
  number = {18},  
  pages  = {18 -- 24}}
```

In text

`\cite{Ciamarra-2005}` → Ciamarra et al. (2005)

List of references:

Ciamarra, M., and Coniglio, A., and Nicodemi, M. (2005). Shear instabilities in granular mixtures. *Physical Review Letters*, vol. 94, no. 18, pp. 18 – 24.



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For a proper working L^AT_EX installation you need the following:

- 1 A L^AT_EX system with programs and packages.
- 2 A L^AT_EX text editor (IDE)
- 3 A BibT_EX bibliographic database manager
- 4 Graphics software.

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The L^AT_EX distributions provide comprehensive T_EX/L^AT_EX systems with binaries for Linux, Windows and Apple Mac OS X. It includes all the major T_EX-related programs, macro packages, and fonts, including support for many languages around the world.

Linux, Windows



Windows



Mac OS X



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TeXstudio a free LaTeX editor

GasDyne - TeXstudio

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 - Dart gun gas dynamics
 - Unsteady flow
 - Dart gun flow
 - Example

```

\item[\subint{e}] exit
\item[\subint{i}] inlet
\item[\RES] reservoir
\item[\VLV] valve

\end{Nomencl}

\clearpage

\section{Dart gun gas dynamics}

\begin{figure}
\centering
\includegraphics[figs/CntrlVol]
\caption{Unsteady flow through a control volume}
\label{fig:CV}
\end{figure}

\subsection{Unsteady flow}
Consider the control volume  $\text{\subint{cv}}$  in  $\text{\figref{fig:CV}}$ . We are only interested in boundary (piston) work on it. The first law of thermodynamics for unsteady flow through a control volume, the rate of change for energy and mass are

\begin{align}
&\frac{dE_{cv}}{dt} = \frac{d(m_{cv}e_{cv})}{dt} \\
&= \dot{m}_{in}e_{in} - \dot{m}_{out}e_{out} + \dot{Q}_{cv} - \dot{W}_{cv} \\
&\frac{dm_{cv}}{dt} = \dot{m}_{in} - \dot{m}_{out}
\end{align}

\begin{align}
&\frac{dE_{cv}}{dt} = \frac{d(m_{cv}e_{cv})}{dt} = \dot{m}_{in}e_{in} - \dot{m}_{out}e_{out} + \dot{Q}_{cv} - \dot{W}_{cv} \\
&\frac{dm_{cv}}{dt} = \dot{m}_{in} - \dot{m}_{out}
\end{align}

with  $\dot{Q}_{cv}$  the net heat input rate and  $\dot{W}_{cv}$  the net work output rate and


$$e = u + \frac{1}{2}c^2 + gz$$


$$\theta = e + Pv = h + \frac{1}{2}c^2 + gz$$


Assume that the process time is fast and that there is no heat transfer,  $\dot{Q}_{cv} = 0$ . We can also ignore the kinetic and potential energy terms. Equation (1) can then be written as


$$\frac{dm_{cv}}{dt}u_{cv} + m_{cv}\frac{du_{cv}}{dt} = \dot{m}_{in}h_{in} - \dot{m}_{out}h_{out} - \dot{W}_{cv}$$


The work performed on the system is for the moving boundary


$$\dot{W}_{cv} = p_{cv}\frac{dV_{cv}}{dt}$$


For an ideal gas we have  $u = C_vT$ ,  $h = C_pT$  and  $\gamma = C_p/C_v$ . Substitute into equation (5)

```

1. Dart gun gas dynamics

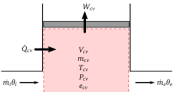


Figure 1: Unsteady flow through a control volume

1.1. Unsteady flow

Consider the control volume (cv) in figure 1. We are only interested in boundary (piston) work on it. The first law of thermodynamics for unsteady flow through a control volume, the rate of change for energy and mass are

$$\frac{dE_{cv}}{dt} = \frac{d(m_{cv}e_{cv})}{dt} = \dot{m}_i\theta_i - \dot{m}_e\theta_e + \dot{Q}_{cv} - \dot{W}_{cv} \quad (1)$$

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e \quad (2)$$

with $\dot{Q}$ the net heat input rate and $\dot{W}$ the net work output rate and

$$e = u + \frac{1}{2}c^2 + gz \quad (3)$$

$$\theta = e + Pv = h + \frac{1}{2}c^2 + gz \quad (4)$$

Assume that the process time is fast and that there is no heat transfer, $\dot{Q} = 0$. We can also ignore the kinetic and potential energy terms. Equation (1) can then be written as

$$\frac{dm_{cv}}{dt}u_{cv} + m_{cv}\frac{du_{cv}}{dt} = \dot{m}_ih_i - \dot{m}_eh_e - \dot{W}_{cv} \quad (5)$$

The work performed on the system is for the moving boundary

$$\dot{W}_{cv} = p_{cv}\frac{dV_{cv}}{dt} \quad (6)$$

For an ideal gas we have $u = C_vT$, $h = C_pT$ and $\gamma = C_p/C_v$. Substitute into equation (5)

JabRef bibliography reference manager

JabRef - /home/alver/jabdoc/main_base.bib

File Edit View BibTeX Tools Web search Plugins Options Help

Groups: tre_entries.bib example_entries.bib main_base.bib

All Entries

- Cod
 - Ingestion
 - Energetics
 - Modelling
- Salmon
 - Modelling
- Rotifer
 - Modelling
 - Enrichment
 - Water quality

Settings

Search

Search All Fields

Clear

☐ Incremental
☒ Float
☐ Filter
☐ Show results in dialog
☐ Global search

Settings

Required fields Optional fields General Abstract Review BibTeX source

#	Entry...	Author	Title	Year	Journal	Time...
73	Article	Breckling et al.	Individual-based models as tools for e...	2006	Ecologic...	2006.0...
74	Article	Brett and Müller-Navarra	The role of highly unsaturated fatty aci...	1997	Freshwa...	
75	Article	Bricaud et al.	Optical-properties of diverse phytopla...	1988	Journal ...	2011.0...
76	Article	Bricaud et al.	Natural variability of phytoplanktonic a...	2004	Journal ...	2011.0...
77	Article	Bricaud et al.	Variations of light absorption by suspe...	1998	Journal ...	2010.1...
78	Article	Bricaud et al.	Absorption by dissolved organic matte...	1981	Limnolo...	2011.0...
79	Article	Browman	Embryology, ethology and ecology of o...	1989	Brain Be...	
80	Article	Browman et al.	Perspectives on ecosystem-based app...	2004	Marine E...	
81	Inbook	Brown and N(\u{x}\~n)es	Fish Diseases and Disorders	1998		2006.0...
82	Article	Brown	Toward a metabolic theory of ecology	2004	Ecology	2008.1...
83	Article	Brown et al.	Larviculture of Atlantic cod (\textit{Gad...	2003	Aquacult...	
84	Article	Brown et al.	The use of behavioural observations in...	1997	Aquacult...	
85	Article	Brown et al.	Nutritional properties of microalgae for...	1997	Aquacult...	2005.1...

Article

Author: Bricaud, A. and Bedhomme, A. L. and Morel, A.

Title: Optical-properties of diverse phytoplanktonic species -- Experimental results and theoretical interpretation

Journal: Journal of Plankton Research

Year: 1988

Volume: 10

Pages: 851--873

Editor:

Bibtexkey: Bricaud1988

Status: Preferences recorded.

UStthesis Class/style files to typeset reports, theses and dissertations that conform to the requirements of Stellenbosch University.

USbib A BibT_EX package for the formatting of bibliographic references of theses.

UStitle A package that redefine the title page to add a logo at the top and an address line below the author.

USnomenc Simple utility to set a nomenclature or list of symbols.

Ussummary Summary page required for the final year projects of the M&M Department.

USlogos A collection of Stellenbosch University crest and logos and Engineering logos.

These packages are all part of the standard T_EXLive and MiK_TE_X distribution. Do not use old versions!



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For more information on $\text{\LaTeX}$:

<https://latex-project.org>
<https://latex-project.org/guides>
<http://www.tug.org>
<http://www.tug.org/interest.html>
<https://www.ctan.org>

Tutorials & wikis

<https://www.ctan.org/pkg/lshort-english>
<https://en.wikibooks.org/wiki/LaTeX>
<https://www.andy-roberts.net/writing/latex>
<http://latex.silmaril.ie/formattinginformation>
<http://tug.org/tutorials/tugindia>
<http://csweb.ucc.ie/~dongen/LAF/LAF.html>

StackExchange Q&A's

<https://tex.stackexchange.com>



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